

An algebraic identity:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2.$$

The computation of the sample variance can also be eased by noting that

$y_i = a + bx_i$, $i = 1, 2, \dots, n$, then $\bar{y} = a + b\bar{x}$, and so

$$\sum_{i=1}^n (y_i - \bar{y})^2 = b^2 \sum_{i=1}^n (x_i - \bar{x})^2.$$

That is, if s_y^2 and s_x^2 are the respective sample variances, then

$$s_y^2 = b^2 s_x^2.$$

Ex: The following data give the worldwide number of fatal airline accidents of commercially scheduled air transports in the years from 1997 to 2005.

Years	1997	1998	1999	2000	2001	2002	2003	2004	2005
Accident	25	20	21	18	13	13	7	9	18

Source: National Safety Council

Find the sample variance of the number of accidents in these years.

Solⁿ: Subtracting 18 from each value, obtain the new data set
7, 2, 3, 0, -5, -5, -11, -9, 0

Calling the transformed data set $y_1, y_2, \dots, y_9$, we have

$$\bar{y} = \frac{1}{9} \sum_{i=1}^9 y_i = -2, \quad \sum_{i=1}^9 y_i^2 = 49 + 4 + 9 + 25 + 25 + 121 + 81 = 314$$

$$s_y^2 = \frac{1}{8} [314 - 9(4)] = 34.75 \quad \therefore s_x^2 = 34.75.$$

Another useful measure of dispersion is the sample range

$$R = \max(x_i) - \min(x_i)$$

It ignores all the information in the sample between the smallest and largest observations. For small sample sizes, say $n \leq 10$, this information loss is not too serious in some situations.)

- A measure of relative variation called the sample coefficient of variation is defined as

$$CV = \frac{s}{\bar{x}}$$

The coefficient of variation is useful when comparing the variability of two or more data sets that differ considerably in the magnitude of the observations.

- Grouped data: If the data are in a frequency distribution, it is necessary to modify the computing formula for the measures of central tendency and dispersion. Suppose that for each of p distinct values of x , say $x_1, x_2, \dots, x_p$, the observed frequency is f_j .

Then the sample mean and sample variance may be computed as

$$\bar{x} = \frac{\sum_{j=1}^p f_j x_j}{\sum_{j=1}^p f_j} = \frac{\sum_{j=1}^p f_j x_j}{n}$$

and

$$s^2 = \frac{\sum_{j=1}^p f_j x_j^2}{n-1} - \frac{1}{n} \left(\sum_{j=1}^p f_j x_j \right)^2$$

respectively.

Measure of Skewness:

$$\frac{m_3}{s^3}$$

Skewness reflects the degree of symmetry about the mean; and negative skewness results from an asymmetric tail toward smaller values of the variables, while positive skewness result from an asymmetric tail extending toward the larger values of the variable.

Measure of kurtosis:

$$\frac{m_4}{s^4} - 3$$

Kurtosis describes the relative peakedness of a distribution as compared to a normal distribution, where a negative value is associated with a relatively flat distribution and a positive value is associated with relatively peaked distributions.